13/20

Blssed G/Michael Catholic Secondary School

1st Semester Mathematics test Two for Grade 10th

Dec, 2018 E.C Time Allowed 1:20'

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I. Choose the best answer among the given alternatives.

C ----1. The Simplest form of $\frac{(-3y)^2}{(-y)^5}$ is :-------

A. 9 B. -9 C. $-9y^{-3}$ D. $\frac{9}{y^3}$

C ----2. If $\log_3 y2 = 2$ then what is the value of y ?

A. 9 B. -9 C. 3 D. 0

C ----3. If $4^x = \frac{1}{16}$ then what is x

A. 1 B. -1 C. -2 D. 2

B ----4. The simplest form of $\sqrt[5]{243}$ is ----------

A. $\frac{1}{3}$ B. 3 C. 5 D. $\frac{1}{5}$

6

C ----5. If $(9^x)\ (4^x) = 6\sqrt{6}$ then x = ----------.

A. 3 B. $^3/_4$ C. $^4/_3$ D. 4

D ----6. For any positive real number x, which one is equal to $(\log_a x)^2$?

A. $2\log_a x$ B. $\log_a x2$ C. $\frac{\log_a x}{2}$ D. $\log_a x \cdot \log_a x$

D ----7. For what value of x is the equation $3^{x-1} = \sqrt{3}$

A. 2 B. $\frac{1}{2}$ C. $^2/_3$ D. $^3/_2$

B ----8. If $\log_a x = \log_x a$ for a positive number x & a then x = ----------

A. a^2 B. $\sqrt{a}$ C. { } D. $\frac{1}{a}$

A ----9. If $3\left(\frac{9}{25}\right)^{x-1} = 5\left(\frac{27}{125}\right)^x$ then x = ----------

A. 0 B. 1 C. 2 D. -1

D ----10. What is the value of x in $\log_2 x + \log_2(x+2) = \log_2 8$ then

A. -4 B. 2 C. 6 D. A & B

II. Give short answer only

11. The simplest form of $^3/_8\left(\log_{1/8} {}^1/_{128}\right)$ is ----------

12. If $\log_4 x - \log_4(x-1) = \log_2 8$ then x = ----------

13. $\log_4 x = {}^1/_2$ then x = ----------

14. What is the simplest form of $\frac{(\sqrt[3]{32})^2}{4^{5/3}}$ = ----------

15. If log 2 = 0.3010 and log 3= 0.4771 then $\log_{\frac{1}{3}} 0.0002$ = ----------

III. Workout

16. What is the simplest form of $\log_4 \sqrt{16}$

17. Simplify $\frac{(n^2)\ (n^3)}{n^{-2}}$

18. If $\log_2(x-1) + \log_2 3 = 3$ then x = ----------

19. log 4 + log 25 = ----------

20. If $\log_x 3 = \frac{1}{3}$ then x = ----------

7X2=14

Soln

16, $\log_4 \sqrt{16} = \log_4 16^{1/2} = \log_4 (4^2)^{1/2} = \log_4 4^{2/2} = \log_4 4 = 1$ //

17, $\frac{(n^2)(n^3)}{n^{-2}} = \frac{n^{2+3}}{n^{-2}} = \frac{n^5}{n^{-2}} = \frac{n^5}{\frac{1}{n^2}} = n^5 \div \frac{1}{n^2} = n^5 * \frac{n^2}{1} = n^{5+2} = n^7$ //

18, $\log_2(x-1) + \log_2 3 = 3 \Rightarrow \log_2 (x-1)(3) = 3 \Rightarrow \log_2 3x-3 = 3$

$= 2^3 = 3x-3 \Rightarrow 8 = 3x-3 = \frac{11}{3} = \frac{3x}{3}$ $x = \frac{11}{3}$ //

19, $\log_{10} 4 + \log_{10} 25 = \log_{10} 4 \times 25 = \log_{10} 100 = \log_{10} 10^2 = 2\log_{10} 10 = 2 * 1 = 2$ //

20, $\log_x 3 = \frac{1}{3} = x^{1/3} = 3 = x = 3^3 = x = 27$ //